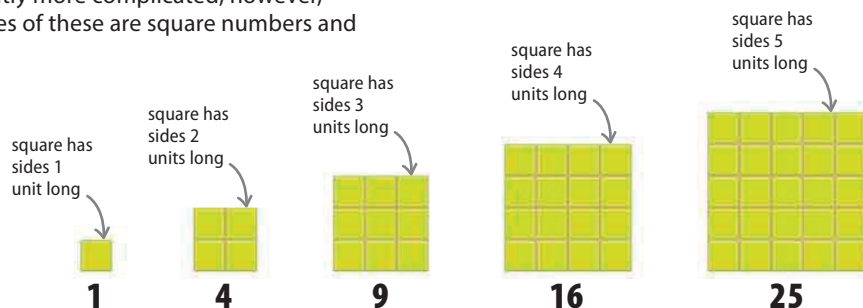


IMPORTANT SEQUENCES

Some sequences have rules that are slightly more complicated; however, they can be very significant. Two examples of these are square numbers and the Fibonacci sequence.

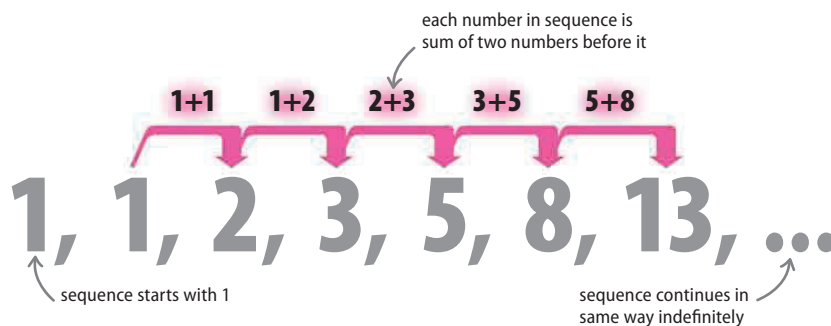
Square numbers

A square number is found by multiplying a whole number by itself. These numbers can be drawn as squares. Each side is the length of a whole number, which is multiplied by itself to make the square number.



Fibonacci sequence

The Fibonacci sequence is a widely recognized sequence, appearing frequently in nature and architecture. The first two terms of the sequence are both 1, then after this each term is the sum of the two terms that came before it.



REAL WORLD

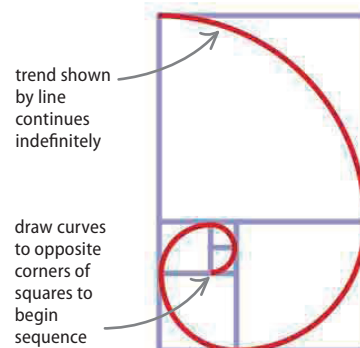
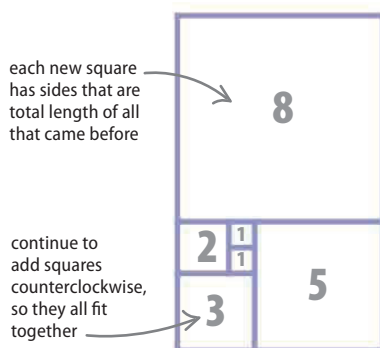
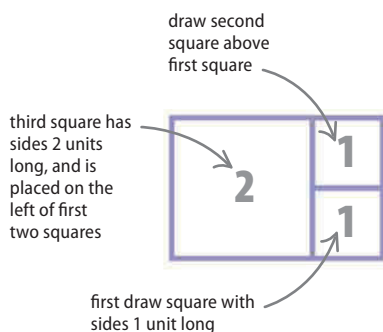
Fibonacci and nature

Evidence of the Fibonacci sequence is found everywhere, including in nature. The sequence forms a spiral (see below) and it can be seen in the spiral of a shell (as shown here) or in the arrangement of the seeds in a sunflower. It is named after Leonardo Fibonacci, an Italian mathematician.



How to draw a Fibonacci spiral

A spiral can be drawn using the numbers in the Fibonacci sequence, by drawing squares with sides as long as each term in the sequence, then drawing a curve to touch the opposite corners of these squares.



First, draw a square that is 1 unit long by 1 unit wide. Draw an identical one above it, then a square with sides 2 units long next to the 1 unit squares. Each square represents a term of the sequence.

Keep drawing squares that represent the terms of the Fibonacci sequence, adding them in a counterclockwise direction. This diagram shows the first six terms of the sequence.

Finally, draw curves to touch the opposite corners of each square, starting at the center and working outward counterclockwise. This curve is a Fibonacci spiral.